Proof of Mind

A Value Chain for Verified Mental Contribution

Will Glynn*
noesis

Version 1.2 June 2026

Abstract

Bitcoin is widely imagined as a value chain: a ledger on which doing valuable work earns provable, accruing, transferable value. It is not. Bitcoin is a *possession* chain. It records who holds which token, orders blocks by burned energy, and lets an external market price the token; the "work" in proof-of-work is decoupled from any useful output. This paper specifies the system the folk myth describes: a chain on which value is *created* as units of contribution, *measured endogenously* by a cooperative-game value function over real outcomes, *owned and transferred* in the manner of unspent transaction outputs, and used to *secure consensus*, by *Proof of Mind* (PoM): a proof of verified, synergy-weighted mental contribution in place of a proof of wasted energy. We give the architecture (provenance-complete blocks, Bitcoin-shaped ownership, a synergy value aggregated by the Myerson value over a provenance graph, elicited by a learned reward model and estimated by permutation sampling), the consensus rule (PoM-weighted, with a participation-aware finalization basis and a coalitional stability constraint), and the mapping from Theory of Mind to Proof of Mind that makes a decentralized network of minds tractable. Because value is measured on-chain and conserved along provenance, the chain is also forwards-compatible: rival useful-work chains can converge into one attribution graph rather than compete for adoption. We are explicit about the one hard problem a value chain cannot sidestep, measuring real value across the blockchain–reality airgap with no external market to price it; we treat it as the load-bearing open problem, bounded by augmented mechanism design rather than claimed solved. The mechanisms reported here are implemented and tested in an open reference node; we mark throughout what is demonstrated and what is designed.

1 Introduction: the value chain Bitcoin is mistaken for

Proof-of-work secures *order*, not *value*. A miner proves it expended energy; the network agrees on a transaction ordering; the coin's worth is set off-chain by a market. Nowhere does the chain record *who contributed what value*. The popular intuition, that miners do work and earn value in proportion to it, is a category error: the work is arbitrary, and the value is exogenous.

A genuine value chain would instead record contribution itself. Each unit of value created, measured by what it actually added, attributed to its creator, transferable, and load-bearing for consensus. The reason such a chain is rare is not ideology but difficulty. A possession chain never has to *measure* value, because the market does; a value chain must. The reason this is hard has a name: the gap between what a chain can verify and what is true in the world, the *blockchain–reality airgap*. Bitcoin sidesteps the airgap by outsourcing valuation to an external market; a value chain has to bridge it, and bridging it objectively and un-gameably is the part this paper takes seriously. The same airgap recurs at every scale of the design: between one mind

*Written with JARVIS, an AI research collaborator. The collaboration is itself an instance of the contribution economy this paper specifies: a human and a machine producing provenance-complete, attributable work.

and another (whom to trust), and between this chain and the chains it would merge (Sections 4 and after).

Proof of Mind is our answer. Its lineage is a single line of descent: proof-of-work, then proof-of-*useful*-work (information density as work), then **proof of mind**, a proof of verified, synergy-weighted contribution, as the apex. The unit of work is a block of thought; the proof is that the thought contributed measurable value. Figure 1 shows how one unit of contribution becomes both consensus weight and tradable state.

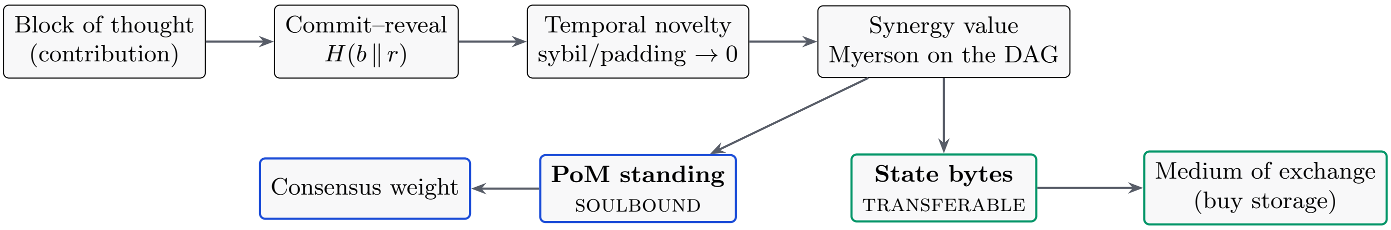

Figure 1: The contribution pipeline. A signed, provenance-complete block is ordered by commit–reveal, priced by temporal novelty, aggregated by a synergy value function, and split into a soulbound standing (consensus weight, cannot be sold) and a transferable state-capacity (a tradable commodity). You can buy storage; you cannot buy consensus.

2 Blocks and provenance

A block is the unit a participant produces, $b = \{\,\mathrm{id},\ \mathrm{parent},\ t,\ \mathrm{inputs},\ \mathrm{output},\ h\,\}$. Inputs are retained so that *how* the output came about is provable and reproducible: provenance, not merely possession. Authorship is made un-front-runnable by **commit–reveal**. A producer publishes a commitment

$$c \;=\; H(b \,\|\, r) \tag{1}$$

with a signature and timestamp *before* revealing the content b and the secret r. This binds authorship and ordering before disclosure, the same mechanism a fair batch exchange uses to eliminate front-running. Committing and then failing to reveal valid content is slashable. Crucially, the commit order is not only an anti-front-running device: it supplies the canonical ordering that makes the value rule of Section 4 strategyproof.

3 Ownership: Bitcoin-shaped, recursive for co-authorship

Each block is locked to an owner public key. Only the current owner's key produces a valid attestation, and ownership is transferable: the owner signs a reassignment to a new key, exactly as a UTXO is spent to a new locking script. Current ownership is not stored as a mutable table; it is *derived* by folding a signed transfer log over a genesis owner,

$$\mathrm{owner}_t \;=\; \mathrm{fold}\big(\mathrm{owner}_0,\ [\,\tau_1, \tau_2, \ldots, \tau_t\,]\big), \tag{2}$$

so there is no ownership record to forge; there is only a signature chain to verify. Transfer voids the prior attestation, and the new owner must re-sign.

A real block often has several authors: a prompt, a generation, a correction. Ownership is then a *share vector*, a set of (contributor, share) pairs, like a multi-output transaction. The within-block shares are computed by the *same value mechanism one level down*: the contributors are players in an intra-block coalition game, and their shares are the Myerson value of their marginal contributions to that block's output. The economy is two-level recursive, outcome → blocks → contributors, and PoM sums a contributor's shares across all blocks.

4 Endogenous value: an honest, synergy-aware price

The central object is a value function $v(S)$ over sets of blocks. Everything else, the score, the stake, the consensus weight, is a functional of v. The difficulty is that the obvious choices of v are either trivial or gameable, and the contribution of this section is a v that is neither.

4.1 Elicitation: from pairwise judgment to cardinal value

Human and model judgments arrive as *pairwise* preferences: block i contributed more than block j. Under the Bradley–Terry model each item has a latent strength s_i and

$$\Pr[\, i \succ j \,] \;=\; \frac{e^{s_i}}{e^{s_i} + e^{s_j}} \;=\; \sigma(s_i - s_j), \tag{3}$$

which is logistic regression on the score difference; the maximum-likelihood fit is convex and unique whenever the comparison graph is strongly connected. Generalized from a fixed set of items to a learned function $r_\theta(x)$ of block features, the same objective

$$\mathcal{L}(\theta) \;=\; -\sum_{(i,j)\,:\, i \succ j} \log \sigma\big(r_\theta(x_i) - r_\theta(x_j)\big) \tag{4}$$

is exactly the loss used to train a reinforcement-learning-from-human-feedback reward model. The distilled judgment model and the reward model are the same object, and that object is the production evaluator for v.

4.2 The additivity trap

A natural first attempt sets a block's value to its share of pairwise wins. We built this, and it fails in an instructive way. If the coalition value is additive, $v(S) = \sum_{i \in S} v(\{i\})$, then the Shapley value collapses to the standalone value, $\phi_i = v(\{i\})$, and the normalized allocation is just the proportional (Copeland) win-share.

Claim 1. *For an additive game the entire $2^{|N|}$-coalition Shapley computation returns the same numbers as a one-line normalization. The cooperative machinery does no work.*

Pairwise comparison yields a *ranking*, which is additive by construction, so Shapley over it is ceremonial. Value measurement is meaningful only when $v(S)$ has **synergy**: a coalition's worth differs from the sum of its members' standalone worth, through complementarity, redundancy, or pivotality.

4.3 Synergy and the Myerson value

We therefore define $v(S)$ as an *outcome value*: the quality or completeness of the outcome reconstructable using only the blocks in S. A block the outcome fails without is pivotal and earns a high marginal value; a block redundant with others earns little. Credit is the **Myerson value**, the Shapley value of the graph-restricted game in which only coalitions connected in the provenance graph G create value:

$$\phi_i(v) \;=\; \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \left[\, v^G(S \cup \{i\}) - v^G(S) \,\right], \qquad v^G(S) = \sum_{T \in S/G} v(T), \tag{5}$$

where S/G are the connected components of S in G. Value flows along provenance edges, not over arbitrary sets; plain Shapley is the special case of a complete graph. Exact computation is exponential, so we estimate ϕ_i by Monte-Carlo permutation sampling (Data-Shapley), which is Shapley specialized to data valuation. In a working prototype over real blocks, replacing the additive game with a submodular coverage outcome-value made the Myerson values diverge materially from the proportional baseline, with mean absolute difference $\|\phi - \text{Copeland}\|_1 \approx 0.26$: the cooperative value became load-bearing, rewarding pivotal blocks and discounting redundant ones.

4.4 Strategyproofness: temporal novelty

A value function over content is gameable by repetition unless novelty is priced. We use the commit order to do exactly that. The value of a block is its coverage *minus* the coverage already claimed by all earlier-committed blocks,

$$\nu(b) \;=\; \Big|\; \mathrm{cov}(b) \;\setminus \bigcup_{c\,:\,t_c<t_b} \mathrm{cov}(c) \;\Big|. \tag{6}$$

First-to-reveal earns the novelty; later padding or sybil copies earn zero. (This is the second reason the commit order of Section 2 is load-bearing: it supplies the canonical order that makes the rule strategyproof.) In the reference implementation a sybil ring of five identical-content identities banks the coverage of at most one block (1.06×, not 5×), and a cross-block re-post earns zero.

4.5 Saturation under coordinated volume

Novelty defeats duplication, but a subtler attack remains: a vested participant, or several colluding ones, can post many distinct-but-low-value children of a valuable parent and try to pump the parent's aggregated flow. We damp this with a single geometric decay over the global, canonically-ordered list of a parent's contributing children. With decay $\rho = 1/\varphi$ (the inverse golden ratio), the parent's aggregated contribution is

$$F \;=\; \sum_{j\geq 0} \rho^j\, c_{(j)} \;\leq\; \frac{c_{\max}}{1-\rho} \;=\; \varphi^2\, c_{\max} \;\approx\; 2.618\, c_{\max}, \tag{7}$$

a convergent series: coordinated volume *saturates* rather than amplifies. An earlier design that damped two axes separately (within an identity and across identities) left their product unbounded, a "diagonal" pump in which K coordinated identities each posting M children compounded both tails. Folding both into one joint decay over the flattened global order closes it. Figure 2 shows the measured effect: before the fix the diagonal grows past the honest single-identity ceiling; after it, the whole grid is bounded by that ceiling.

4.6 A worked example

A small example ties the pieces together. Take a result A with coverage $\{1,2,3\}$ committed first, a block B built on A with coverage $\{3,4\}$ committed second, and a near-duplicate D of B committed third. Temporal novelty credits coverage only against everything committed earlier: $\nu(A) = |\{1,2,3\}| = 3$, $\nu(B) = |\{4\}| = 1$ (shingle 3 was already claimed by A), and $\nu(D) = 0$ (its coverage is wholly claimed). Padding and sybil copies earn nothing, concretely.

For credit, let the outcome value of a connected set be its covered shingles, $v(S) = \big|\bigcup_{i\in S}\mathrm{cov}(i)\big|$, so $v(\{A\}) = 3$, $v(\{B\}) = 2$, and $v(\{A,B\}) = |\{1,2,3,4\}| = 4$. The Myerson value over the edge $A - B$ is $\phi_A = \frac{1}{2}(3-0) + \frac{1}{2}(4-2) = 2.5$ and $\phi_B = \frac{1}{2}(2-0) + \frac{1}{2}(4-3) = 1.5$, summing to $v(\{A,B\}) = 4$. A proportional split by standalone value would instead give $A = 2.4$, $B = 1.6$: the synergy rule shifts 0.1 of credit *to* the pivotal A (which owns the shared shingle 3) and away from the partly-redundant B. The cooperative machinery is doing work, not ceremony.

Finally, suppose an attacker floods A with N low-value children each contributing c. The geometric decay above bounds the parent's aggregated flow at $\sum_{j\geq 0}\rho^j c = \varphi^2 c \approx 2.618\, c$ for *any* N: volume cannot pump the gate.

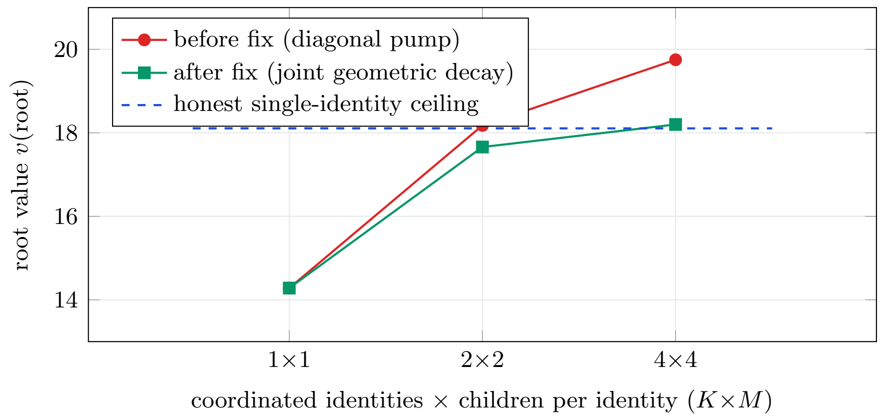

Figure 2: Coordinated-volume saturation, measured in the reference node. Before the joint-decay fix, K coordinated identities each posting M valueless children pump the root's aggregated value past the honest single-identity ceiling (red). After folding the two damping axes into one geometric decay over the global canonical order (green), the entire $K{\times}M$ grid is bounded by that ceiling to within 2%. Honest single contributions are paid unchanged.

4.7 Certifying the value: the HodgeRank residual

Pairwise judgments need not be transitive; intransitivity is inherent (Condorcet), not merely noise. A scalar value function is the right object only when the judgments are close to a gradient flow. We make this checkable. Encoding comparisons as a skew-symmetric edge flow Y on the comparison graph and solving the weighted least-squares problem

$$\min_{s} \sum_{i,j} w_{ij} \left(s_i - s_j - Y_{ij}\right)^2 \tag{8}$$

yields a scalar ranking s (a graph-Laplacian solve, generalizing Borda). The Helmholtz–Hodge decomposition splits the flow orthogonally,

$$Y = \underbrace{\operatorname{grad} s}_{\text{global ranking}} \oplus \underbrace{\ker \Delta_1}_{\text{harmonic (global cycles)}} \oplus \underbrace{\operatorname{curl}^*}_{\text{local cycles}} . \tag{9}$$

The residual energy is not an error bar; it is a *certificate*. A large curl component says fine-scale ordering is unreliable; a large harmonic component says the judgments cycle through holes in the graph. Where the Shapley/Myerson construction *assumes* a consistent scalar world (its axioms force the residual to zero), HodgeRank *measures* whether that world holds, and localizes where it breaks. This matters for security as well as honesty: a coordinated sybil or collusive ring injects detectable harmonic circulation, so a spike in harmonic energy is a manipulation alarm in the same computation that produces the value. The value layer therefore ships its own certificate of when the single number it reports can be trusted. A large residual is not merely reported: it routes the contested credit away from the scalar to a higher-cost path, a multi-dimensional split or a human jury, spent only where the judgments actually disagree.

4.8 The learned evaluator, bounded by construction

The production $v(S)$ is a learned reward model, and learned models drift and can be probed. We do not ask it to be un-gameable. We deny it the authority to be dangerous. The strategyproof skeleton is the realized gate

$$\text{seed}_i = \bar{\nu}_i \cdot g(f_i) \cdot \omega(\ell_i), \qquad \bar{\nu}_i,\ g(f_i),\ \omega(\ell_i) \in [0,1], \tag{10}$$

the product of floored novelty $\bar{\nu}_i$, a realized-flow gate $g(f_i)$, and the learned outcome ω scored on the block's provenance lineage ℓ_i. Because the factors are multiplicative and bounded in $[0, 1]$, each can only *lower* the seed; none can mint. The learned model is confined to two roles that have no minting power: it may offer an honest contributor an early liquidity *advance* against expected vesting (its own capital at risk, repaid from realized value and slashed on shortfall), and it may enter a dispute as *evidence* the jurors weigh. A corrupt model that scores everything maximally reduces to the un-learned gate exactly: in the reference node, $v_8 \equiv v_7$ under an adversarial $+\infty$ model. The proof obligation collapses from "the neural network is un-gameable" to "three bounds hold," which are small functions with tests.

The value of the learned model is what it adds *within* those bounds. Figure 3 reports a held-out measurement: trained on one split of coalitions and tested on coalitions never seen, the learned $v(S)$ ranks "work built on work" above the same cells dumped as orphans with accuracy ≥ 0.9, while the coverage proxy is blind to lineage and ties at chance.

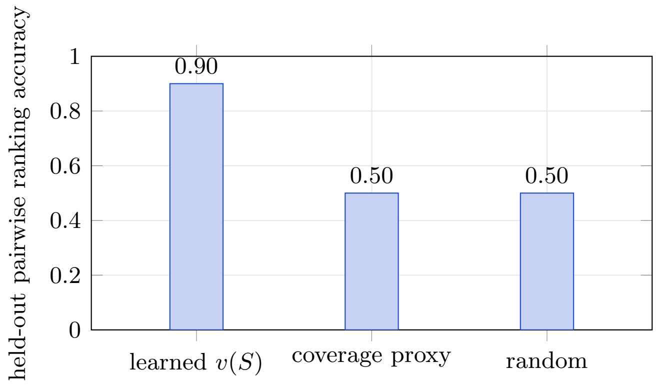

Figure 3: The moat, measured not asserted. On coalitions never seen in training, the learned $v(S)$ distinguishes connected, synergistic contribution from the same content as disconnected orphans (≥ 0.9), while the coverage proxy the per-block rule can see is blind to lineage and ranks at chance (0.5). The remaining mile is real outcome-preference data; the measurement harness runs unchanged when it lands.

5 Measurement as a living mechanism

The value function of Section 4 is not a fixed formula, and it cannot be one. Any static value rule is Goodhart-bait: the moment it is published, it is gamed, because a fixed objective tells an adversary exactly what to optimize. The only un-gameable measure is one that adapts to the gaming. Measurement on this chain is therefore a closed loop, not a closed form, and its components, introduced separately above, are one adaptive system.

1. **Realized, not predicted.** The learned model can only lower a seed it never raises one (Section 4). Value is gated on realized downstream flow, so the measure is anchored to what actually happened, not to a prediction an adversary can spoof.

2. **It retrains on outcomes.** The chain is a clean, provenance-complete, value-weighted dataset (Section 9). The evaluator is fine-tuned on verified high-value contributions against caught failures, so the measure sharpens as the chain accumulates verified truth. Yesterday's attack becomes today's training signal.

3. **Value is provisional, then adjudicated.** A payout is not final at emission. A challenge window lets a bonded challenger refute a value claim; a refuted claim is zeroed and the claimant slashed. Measurement is a dispute process with finality, not a one-shot score.

4. **It cannot be banked.** Standing decays, so a high score earned once does not persist without continued contribution. The measure tracks live value, and a stale position is reclaimed.
5. **It diagnoses its own breakdown.** The HodgeRank residual (Section 4) is a continuous detector: when the judgments stop fitting a single scalar, through genuine multidimensionality or through a coordinated manipulation injecting harmonic circulation, the residual energy rises and flags it. The system therefore knows *when* to distrust its own number, rather than reporting a confident scalar over a structure that no longer supports one.
6. **The dimensions of value can themselves evolve.** The value matrix is governance-mutable but verifier-gated: a new dimension is admitted only if it predicts realized downstream value, weights move within bounded ranges, and no real dimension can be zeroed. The measure is not frozen at genesis, and it cannot be captured into a private one.

Put together, the loop is: measure, realize, dispute, retrain, decay, re-measure. A static rule loses to Goodhart on its first day; an adaptive measure with an un-gameable realized-value anchor, a dispute process for finality, and a residual that reports its own validity does not present a fixed target to optimize against. This is why Section 10 calls value measurement the load-bearing open problem rather than a solved one. The claim is not a closed-form proof of un-gameability; it is a mechanism that treats un-gameability as a moving equilibrium it must continuously hold, and whose present strength is measured (Figure 3) rather than asserted. A static rule, exact on paper but gameable in practice, is wrong in the only sense that matters; an adaptive measure that is approximately right and provably converging is the better object. It is better to be approximately right than absolutely wrong.

6 Proof of Mind

A participant's **PoM score** is its accumulated Myerson value over the verified, owned, provenance-complete blocks it holds,

$$\mathrm{PoM}(a) \;=\; \sum_{b\,\in\,\mathcal{B}(a)} \phi_b, \qquad \mathcal{B}(a) = \{\text{verified, owned, provenance-complete blocks of } a\}. \tag{11}$$

Three properties are inherited from the construction. It is **verifiable**: every contributing block is signed and provenance-complete, so anyone can recompute the score from public data. It is **sybil-resistant structurally**: PoM requires owned blocks whose value was synergy-judged, and splitting one mind across many accounts does not multiply PoM because the synergy game discounts redundant copies; a thousand empty nodes score zero. And it is **earned, not bought**: the stake is accumulated proof of contribution, and slashing revokes PoM for refuted blocks, caught hallucinations, and failed attestations.

6.1 Theory of Mind to Proof of Mind

Across agents, Theory of Mind is inference. You cannot see inside another mind, so you guess whether to trust it. That guess is an airgap, the inter-mind version of the gap between a blockchain and reality. Economic Theory of Mind reframes a mind as an economy whose outputs carry value. Proof of Mind is the verifiable economic proof of that economy, and it closes the airgap: instead of each node inferring whether another is a mind worth trusting (intractable and game-able), PoM supplies a proof anyone can recompute. Trust stops being a per-mind inference and becomes a structural property. The chain is capacity $\to$ mind-as-economy $\to$ proof of that economy.

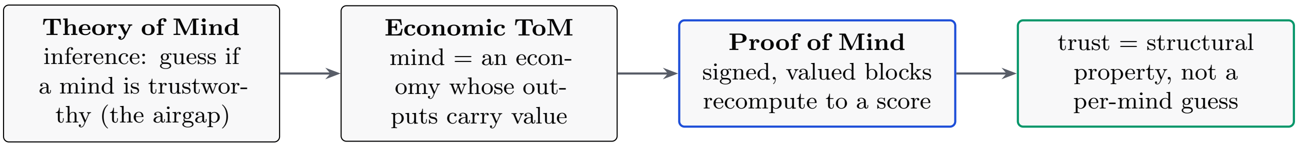

Figure 4: The mapping that makes a decentralized network of minds tractable: replace "infer whether to trust" with "verify a proof."

7 Consensus

Validators are weighted by **PoM**, not by energy or capital. The tamper-evident, signed, owned chain is the ledger; PoM is the stake; consensus is PoM-weighted agreement on the canonical chain. Three design choices make this safe.

7.1 Retention decay and the finalization basis

Vote weight decays with staleness, so standing tracks live participation rather than past glory. Decay applies to the *franchise*, the vote weight, never to the staked balance, and symmetrically across all dimensions so the effective mix is time-invariant. The remaining question is what a supermajority is a fraction *of*. A base-weight denominator is eclipse-resistant but halts under low participation; a pure effective-weight denominator removes the halt but lets an attacker shrink the denominator and finalize a minority. The stable choice is the hybrid

$$\text{finalize} \iff \forall d: \quad W^d_{\text{for}} \;\geq\; \tfrac{2}{3}\,\max\big(W^d_{\text{eff}},\, Q^d\big), \tag{12}$$

which pins the denominator at a quorum floor Q^d, a fraction of base weight, so neither failure occurs. The composition over the dimensions d is AND-gated, not a substitutable weighted sum: a single dimension below the supermajority cannot finalize alone. This is verified against tested reference models, including a self-audit that demonstrates the eclipse on a bare effective basis and its closure under the quorum floor.

7.2 Stability: no profitable fork

PoM-weighted agreement must be defection-proof; no coalition of validators should profit by deviating. We impose a coalitional stability constraint over the PoM-weighted game: require the allocation to lie in the **core** when it is non-empty (no sub-coalition does better by forking), and fall back to the **nucleolus** when the core is empty (the unique allocation that lexicographically minimizes the worst coalition's incentive to defect). This is added precisely because consensus requires it; pure attribution does not need a stability concept.

7.3 Three powers

Consensus is not a single weighted vote. It is the agreement of three independent powers, **cognition** (PoM), **capital** (state-stake), and **compute** (proof-of-work, carried by a separate money layer), composed by conjunction. Three is the minimum for a non-dominated cyclic equilibrium: with two powers one is the strict best response and captures the other; with three the relation is non-transitive (rock–paper–scissors) and capture-resistant; four or more adds complexity without more non-domination. Because the composition is an AND of independent layers rather than a weighted sum, a large share in one dimension is a reward weight, not a vote that can finalize alone. An attacker must defeat all three.

8 Cryptoeconomics: one PoM, one byte

Storage is the scarce resource, and PoM is the right to occupy it: **one PoM equals one byte of on-chain state**. Issuance is earned, not bought, because novel verified contribution is the mint; state-rent is PoM decay, which both reclaims idle capacity and serves as the supply sink that bounds total live PoM. The mint (novel contribution) and the burn (decay) balance. A floor protects a genuine contributor from being zeroed by a brief absence: decay reclaims idle capacity, it does not erase earned standing to nothing.

The subtlety that keeps this from collapsing into proof-of-stake is a split between two cells. A participant's **standing** is soulbound: it is the franchise (consensus weight and the right to mint state), and it cannot be sold. The **state-capacity** it earns the right to mint is transferable: a byte of state is a liquid commodity that trades freely. The result is that you can buy storage but not consensus. A transferable franchise would let a rich actor buy consensus, which is proof-of-stake by another name; a soulbound franchise makes "earned, not bought" and structural sybil-resistance real at once.

9 Backwards-enforcement of the model

The value chain is also a training signal. Each block is provenance-complete, owner-authenticated, and Myerson-valued: a clean, value-weighted dataset, exactly the object Data-Shapley attributes. With open model weights, fine-tuning on high-PoM verified blocks against caught hallucinations lets the governance layer's accumulated, verified truth shape the model. With closed weights, the same truth constrains the model in-context: gates block disallowed actions and the signed chain contradicts a hallucination, forcing correction. Governance produces a training signal, which shapes the model, which governance then verifies, and the loop compounds. The structure does not only constrain the mind; it teaches it.

10 Security and honest limitations

10.1 Threat model

The mechanisms above compose into one defense surface. Each attack is matched to the structural property that defeats it and its status in the reference node.

Attack	What defeats it (structural)	Status
Sybil split (one mind, many accounts)	synergy game discounts redundant copies; PoM does not multiply	demonstrated ($1.06\times$ at $K{=}5$)
Padding / re-post of coverage	temporal novelty: later copies earn 0 novel coverage	demonstrated
Coordinated-volume / diagonal pump	single joint geometric decay, bound $\varphi^2 c$ (Fig. 2)	demonstrated
Gamed learned evaluator	bounded role: can only lower a seed, never mint ($v_8 \equiv v_7$ under $+\infty$)	demonstrated
Cyclic / collusive judgments	HodgeRank harmonic-energy spike is a manipulation alarm	designed
Self-referential scoring	independent evaluators, eigenvector damping, synergy discount	designed
Eclipse (shrink the finality denominator)	hybrid max(effective, quorum-floor) basis	demonstrated (self-audit)
Profitable fork (coalition deviates)	core / nucleolus stability constraint	demonstrated (solver)
Spend another owner's cell	existence check + lock-signature control	designed (digest built)
Theft by forking the network	attribution *is* the mechanism: stripping provenance is incoherent, keeping it credits the source	structural
Off-chain sale of a standing key	soulbound blocks on-chain transfer; the residual is selling the identity key itself, bounded because the key inherits the identity's full slashable history (and personhood-binding)	designed (residual)

The load-bearing bet. A value chain must measure value un-gameably across the blockchain–reality airgap (Section 4), the gap between what the chain can verify and what is true in the world. This is the problem Bitcoin sidesteps by being mere possession, letting an external market bridge the airgap for it. If the learned outcome evaluator is gameable, PoM degrades into a reputation system. This is the central open problem, and we treat it as such. The mitigations are structural: the realized-flow gate that the learned model can only lower, the temporal-novelty rule, the HodgeRank manipulation certificate, and the held-out measurement of Figure 3. The remaining mile is real outcome-preference data at scale.

Augmented mechanism design: the counterweight. The concession is bounded, not open-ended, and the bound is the methodology. We do not replace the cooperative-game mechanism with a learned oracle, nor do we trust the oracle; we *augment* the mechanism with invariants that hold by construction. The learned evaluator can only lower a seed, never mint one; value is gated on realized flow; supply is conserved; finality is an AND over independent dimensions. Each invariant is deterministic, so the blast radius of a wrong measurement is bounded *a priori*: at worst the chain under-credits honest work, it never forges value from a gamed score. Within those invariants the living-measurement loop (retrain on outcomes, dispute, decay, residual self-diagnosis) is a deterministic path that converges toward the true measurement as the chain accumulates verified outcomes. The open problem is therefore the distance between *invariant-bounded and converging* and *closed-form-proven*, not between working and not working. This is augmented mechanism design: augment a sound mechanism with math-enforced invariants rather than betting the system on a component being perfect, so the honest concession and a credible deterministic path coexist.

Self-reference. A block can carry information about contributions and thereby influence the scoring of others. The value layer scores such a block for its own marginal contribution; the elicitation layer consumes its content as a judgment about referenced blocks. These must stay decoupled, and no block may be the sole scorer of its own value. Guards: independent evaluators, eigenvector-style damping over the attribution graph, and the synergy game's discounting of self-referential coalitions.

Demonstrated versus designed. *Demonstrated* in the open reference node (a Rust implementation with a no-std verification core shared with on-chain scripts, 281 passing tests at the time of writing): ownership and transfer, per-block signing, tamper-resistance, the synergy value over a coverage proxy with Myerson and Bradley–Terry and permutation sampling, temporal novelty, coordinated-volume saturation (Figure 2), the bounded learned evaluator and its held-out measurement (Figure 3), the dispute and slashing path, PoM-weighted finalization with the hybrid basis, the core/nucleolus stability solver, and the on-chain recomputation of the ordering and finalization rules inside the virtual machine. *Designed, not yet built*: the learned $v(S)$ on real outcome labels, the production two-level recursion, owner-signature control of spends (the deterministic transaction digest the signature will cover is built and tested), the open-weight fine-tune loop, and the money/stability layer.

11 Fair launch and theft-resistance

The creator's pre-launch contribution would otherwise be an unearned head start. It is neutralized by a **genesis-burn**: the chain stays continuous from genesis and pre-launch blocks remain auditable, but their standing and state-value are programmatically burned to zero at the launch height. This is a fair launch one can verify on-chain, chosen over a chain reset, which only asserts fairness.

The mechanism also yields an unusual theft-resistance. Because the system *is* attribution, a derivative deployment has two options. It can strip the provenance, in which case the attribution layer the system depends on is gone and the fork is incoherent; or it can preserve the provenance, in which case its edges trace back to this origin and running it credits the source. Copying the network honestly adds the copier to the same attribution graph, so there is no outside to fork to. Forking, which fragments value on a possession chain, here becomes contribution.

12 Forwards-compatibility: convergence, not competition

A possession chain fragments. Every new design competes to be the one chain, and value splits across the survivors. A value chain can do the opposite. Because contribution here is measured endogenously and conserved along provenance, a contribution from another chain can be imported as an attributed cell, re-scored by the same value function, and credited on one canonical graph without loss. The chain is not only backwards-compatible (continuous from genesis, prior blocks auditable); it is forwards-compatible, in that other useful-work chains can converge into it by a reverse fork. A normal fork diverges, splitting one chain's state into two; a reverse fork converges, merging many chains' contributions into one attribution graph.

This reframes the open question that hangs over every useful-work chain. The question "which one do we adopt" presumes a competition for a single standard. The answer is all of them. The useful work of Primecoin, of a storage market, of a decentralized training run, is in each case a contribution with provenance; a chain that ingests foreign contributions agnostically, scores them by one value function, and credits them on one canonical graph does not have to win the adoption fight, it absorbs it. This is how a fractured ecosystem un-fractures: not by forcing a million chains to adopt one of them, which is impossible, but by giving them a substrate they converge into without anyone losing what they built. The cooperation is structural, a property of conserved contribution, not a treaty that can be defected from.

The enabling property is precisely the endogenous measurement of Section 4. A possession chain cannot losslessly merge another chain because its value is exogenous: priced off-chain by a market, it has no on-chain quantity to carry across the merge. Bitcoin has no organ for valuing Primecoin's primes. A value chain has exactly that organ, and so convergence is the ecosystem-scale expression of the same property that makes the single chain novel. It also generalizes the theft-resistance of Section 11: where that argument left a fork no outside to escape *to*, this leaves a rival chain no outside to converge *from*, since its contributions either join this graph or strip the provenance that gave them worth.

This is a design direction, not a built feature, and it is honest about its hard parts. A chain-agnostic contribution adapter must map a foreign unit to a provenance-bearing cell; cross-chain novelty must hold, so the same contribution arriving from two sources is credited once; and the import boundary is an airgap, so a foreign chain's attestation is bonded and disputed, never trusted, and re-measured by the same value function rather than imported at its source's claimed worth. The merge conserves; it does not import another chain's trust assumptions.

13 Related work and what is novel

Three families are adjacent, and the distinction in each case is the same: the consensus object is something other than an endogenously-measured contribution-value.

Useful proof-of-work. Primecoin, Permacoin, proof-of-replication (Filecoin), proof of space-time (Chia), and the theoretical line of Ball, Rosen, Sabin and Vasudevan make the *work* useful, but value stays exogenous: security comes from puzzle hardness, dedicated space, or honest execution, while the work's worth is priced by an external market or fixed by an external problem-provider. The theoretical line is weaker than it appears (the usefulness claims of the 2017 construction were retracted in 2018), and proof-of-learning schemes are provably not robust to spoofing. Decentralized-training systems (Gensyn, Prime Intellect, Nous) make verified compute or training-integrity the consensus object, not output value. Decentralized-inference networks come closer, and we treat them next.

Output quality by peer agreement (Bittensor, Fortytwo). The closest operating systems do make output quality the reward object, and they are the strongest prior art. Bittensor's Yuma consensus takes a stake-weighted, clipped aggregate of validators' quality scores; Fortytwo aggregates pseudo-random pairwise comparisons among nodes with a Bradley–Terry model, the highest-quality response winning consensus and reputation tracking how often a node's outputs win. The differentiator is *how* the quality is obtained. Both rest on *subjective peer or validator agreement*: Yuma rewards agreeing with the stake-weighted majority, and empirically reward is dominated by purchased stake rather than by quality; Fortytwo's score is a peer ranking, not a value the protocol computes. Neither uses a cooperative-game synergy value over a provenance graph, neither binds the score to a learned value-model with a residual trust-certificate, and in Fortytwo reputation weights consensus without being the finality object itself. Noesis measures value objectively, by construction, and makes that measurement the finality object directly.

Contribution attribution. Ethereum's Deep Funding is the most superficially similar prior work, sharing the "value as a graph" framing and an AI-distilled credit signal. It is nonetheless distinct on two independent axes. It computes a distilled *scalar edge-weight* (the fraction of a dependent's credit assigned to a dependency), regressed by a model to match jury pairwise preferences; it is not a cooperative-game value, has no characteristic function, no coalition marginal contribution, and no connected-coalition restriction. And it is an off-chain *funding-allocation oracle* that splits a fixed pot, never a consensus, finality, or stake-weight object. Data-Shapley applies the flat (non-graph) Shapley value to training data as an offline metric; the Myerson value we use is the graph-restricted generalization (Myerson, 1977), a known primitive whose novelty here is its *application as the consensus object*, not the primitive itself. Every on-chain use of the Shapley value we found redistributes revenue *after* consensus rather than securing it.

Non-transferable standing. SourceCred separates non-transferable earned reputation (Cred, computed by PageRank) from a transferable token (Grain), and the Decentralized Society proposal introduces non-transferable soulbound tokens for reputation and credentials. The split pattern therefore has prior art. Neither, however, makes non-transferable earned standing the *consensus weight* while keeping a separate *transferable state-capacity* token: SourceCred's consensus is social, and the soulbound-token proposal addresses correlation-discounted DAO voting, not a consensus franchise.

Net. *Objective* value measurement as the consensus object, a cooperative-game value over a provenance graph made the finality weight directly (Section 4), is the novel core: the adjacent systems above reward output quality by subjective peer agreement, not by an objective measure. The Myerson-over-provenance-DAG credit rule and the soulbound-standing / transferable-capacity split are novel in their consensus application, though each composes known primitives. The HodgeRank residual as a stake-slashing manipulation alarm is *not* new: Delegated Proof of Reputation (2019) already proposes a Hodge-subspace filter that slashes flows in the harmonic and curl subspaces; our narrower contribution pairs the residual with a learned value-model and the endogenous value object, which that proposal lacks. We state these as positioned claims open to refutation, in the spirit of Section 10.

14 Conclusion

We have specified a chain on which value is created as contribution, measured endogenously by a synergy-aware cooperative value over a provenance graph, owned and transferred like a UTXO, and used to secure consensus through Proof of Mind. The design replaces the inferences that make decentralized trust intractable, "is this a mind worth trusting," with proofs anyone can recompute, and it replaces proof of wasted energy with proof of verified thought. The honest center of the system is a single hard problem, un-gameable value measurement, and the architecture is built so that the learned component charged with it can only ever deny value, never mint it, while a separate certificate reports when the value can be trusted at all. What remains is to feed that evaluator real outcomes at scale. The rest is assembled from known mathematics, and it runs.

Notation

Symbol	Meaning
b	a block: $\{\mathrm{id}, \mathrm{parent}, t, \mathrm{inputs}, \mathrm{output}, h\}$
$H(b \,\\|\, r),\ r$	commitment hash; reveal secret
$v(S)$	outcome value of a set S of blocks (the central object)
ϕ_i	Myerson value (synergy credit) of block / contributor i
$\nu(b)$	temporal novelty of b (coverage not claimed by an earlier block)
$\sigma,\ s_i$	logistic function; latent Bradley–Terry strength of i
r_θ	learned reward model (the production evaluator for v)
$\mathrm{seed}_i,\ \omega(\ell_i),\ \ell_i$	value-gate seed; learned outcome factor; provenance lineage of i
$\rho,\ \varphi$	saturation decay $\rho = 1/\varphi$; golden ratio $\varphi \approx 1.618$
$\mathrm{PoM}(a)$	Proof-of-Mind score of agent a (sum of ϕ over its owned blocks)
W^d_{for}, W^d_{eff}, Q^d	for- / effective- / quorum-floor vote weight in dimension d

(φ, bare, is the golden ratio in the saturation bound; ϕ_i, subscripted, is the Myerson credit. They are distinct.)

References

1. S. Nakamoto. *Bitcoin: A Peer-to-Peer Electronic Cash System*. 2008.
2. L. S. Shapley. *A Value for n-Person Games*. Contributions to the Theory of Games II, 1953.
3. R. B. Myerson. *Graphs and Cooperation in Games*. Mathematics of Operations Research, 1977.
4. A. Ghorbani, J. Zou. *Data Shapley: Equitable Valuation of Data for Machine Learning*. ICML 2019.
5. R. A. Bradley, M. E. Terry. *Rank Analysis of Incomplete Block Designs*. Biometrika, 1952.
6. X. Jiang, L.-H. Lim, Y. Yao, Y. Ye. *Statistical Ranking and Combinatorial Hodge Theory*. Mathematical Programming Ser. B 127, 2011.
7. D. Schmeidler. *The Nucleolus of a Characteristic Function Game*. SIAM J. Applied Math, 1969.
8. L. Gao, J. Schulman, J. Hilton. *Scaling Laws for Reward Model Overoptimization*. 2023.
9. Nervos CKB. *The Common Knowledge Base: a cell-model, state-rent blockchain*. 2019.
10. Ethereum Foundation. *Deep Funding: distilled pairwise judgment over a dependency graph*. 2024.
11. Bittensor / Opentensor. *Yuma Consensus*. 2021.
12. *Fortytwo*. arXiv:2510.24801, 2025.
13. *Delegated Proof of Reputation: a novel blockchain consensus*. arXiv:1912.04065, 2019.